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Asymmetric Battery Having a Semi-Solid Cathode and High Energy Density Anode

Abstract

Embodiments described herein relate generally to devices, systems and methods of producing high energy density batteries having a semi-solid cathode that is thicker than the anode, An electrochemical cell can include a positive electrode current collector, a negative electrode current collector and an ion-permeable membrane disposed between the positive electrode current collector and the negative electrode current collector. The ion-permeable membrane is spaced a first distance from the positive electrode current collector and at least partially defines a positive electroactive zone. The ion-permeable membrane is spaced a second distance from the negative electrode current collector and at least partially defines a negative electroactive zone. The second distance is less than the first distance. A semi-solid cathode that includes a suspension of an active material and a conductive material in a non-aqueous liquid electrolyte is disposed in the positive electroactive zone, and an anode is disposed in the negative electroactive zone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority and benefit of U.S. Provisional Application No. 61/787,372, filed Mar. 15, 2013 and entitled “Asymmetric Battery Having a Semi-solid Cathode and High Energy Density Anode”, the entire disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

[0002] Embodiments described herein relate generally to asymmetric batteries having high energy density anodes, and more particularly to devices, systems and methods of producing high energy density batteries having a semi-solid cathode that is thicker than the anode.

[0003] Batteries are typically constructed of solid electrodes, separators, electrolyte, and ancillary components such as, for example, packaging, thermal management, cell balancing, consolidation of electrical current carriers into terminals, and/or other such components. These conventional battery manufacturing methods generally involve complicated and expensive manufacturing steps, such as casting the electrode, and are only suitable for electrodes of limited thickness, e.g., typically less than 100 μm . These known methods for producing electrodes of limited thickness result in batteries with lower capacity, lower energy density, and a high ratio of inactive components to active material. Said another way, the non-energy storage elements of the finished device, that is the separator and current collector, comprise a relatively high fixed volume or mass fraction of the device, thereby decreasing the device's overall energy density.

[0004] Thus, it is an enduring goal of energy storage systems development to simplify and reduce manufacturing cost, reduce inactive components in the electrodes and finished batteries, and increase energy density and overall performance.

SUMMARY

[0005] Embodiments described herein relate generally to devices, systems and methods of producing high energy density batteries having a semi-solid cathode that is thicker than the anode: In some embodiments, an electrochemical cell can include a positive electrode current collector, a negative electrode current collector and an ion-permeable membrane disposed between the positive electrode current collector and the negative electrode current collector. The ion-permeable membrane is spaced a first distance from the positive electrode current collector and at least partially defines a positive electroactive zone. The ion-permeable membrane is spaced a second distance from the negative electrode current collector and at least partially defines a negative electroactive zone. The second distance is less than the first distance. A semi-solid cathode that includes a suspension of an active material and a conductive material in a non-aqueous liquid electrolyte is disposed in the positive electroactive zone, and an anode is disposed in the negative electroactive zone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a schematic illustration of an electrochemical cell according to an embodiment.

[0007] FIG. 2 is a schematic illustration of a symmetric electrochemical cell, according to an embodiment.

[0008] FIG. 3 is a schematic illustration of an asymmetric electrochemical cell, according to an embodiment.

[0009] FIG. 4 is a schematic illustration of an asymmetric electrochemical cell, according to an embodiment.

DETAILED DESCRIPTION

[0010] Embodiments described herein relate generally to devices, systems and methods of producing high energy density batteries having a semi-solid cathode that is thicker than the anode. In some embodiments, an electrochemical cell can include a positive electrode current collector, a negative electrode current collector and an ion-permeable membrane disposed between the positive electrode current collector and the negative electrode current collector. The ion-permeable membrane is spaced a first distance from the positive electrode current collector and at least partially defines a positive electroactive zone. The ion-permeable membrane is spaced a second distance from the negative electrode current collector and at least partially defines a negative electroactive zone. The second distance is less than the first distance. A semi-solid cathode that includes a suspension of an active material and a conductive material in a non-aqueous liquid electrolyte is disposed in the positive electroactive zone, and an anode is disposed in the negative electroactive zone.

[0011] Consumer electronic batteries have gradually increased in energy density with the progress of lithium-ion battery technology. The stored energy or charge capacity of a manufactured battery is a function of: (1) the inherent charge capacity of the active material (mAh/g), (2) the volume of the electrodes (cm³) (i.e., the product of the electrode thickness, electrode area, and number of layers (stacks), and (3) the loading of active material in the electrode media (e.g., grams of active material per cm³ of electrode media). Therefore, to enhance commercial appeal (e.g., increased energy density and decreased cost), it is generally desirable to increase the areal charge capacity (mAh/cm²). The areal charge capacity can be increased, for example, by utilizing active materials that have a higher inherent charge capacity, increasing relative percentage of charge storing material (i.e., "loading") in the overall electrode formulation, and/or increasing the relative percentage of electrode material used in any given battery form factor. Said another way, increasing the ratio of active charge storing components (e.g. the electrodes) to inactive

components (e.g., the separators and current collectors), increases the overall energy density of the battery by eliminating or reducing components that are not contributing to the overall performance of the battery. One way to accomplish increasing the areal charge capacity, and therefore reducing the relative percentage of inactive components, is by increasing the thickness of the electrodes.

[0012] In some embodiments, higher energy densities and capacities can be achieved by, for example, improvements in the materials used in the anode and/or cathode, and/or increasing the thickness of the anode/cathode (i.e., higher ratio of active materials to inactive materials). One of the latest materials used in the anode for consumer electronics is, for example, silicon (Si), tin (Sn), silicon alloys, or tin alloys due to their high capacity and low voltage. Typically, this high capacity active material is mixed with graphite due to its high first charge capacity and related first charge irreversible capacity. Silicon has a first charge theoretical capacity of 4,200 mAh/g and an irreversible capacity of more than 300 mAh/g. Therefore, typical anodes that utilize Si contain a mixture of silicon and graphite in order to reduce the irreversible capacity. In addition, silicon undergoes a very large volume change during lithium insertion causing the volume of the material to grow by more than 300%. To limit this large volumetric expansion, current high capacity anodes utilize between 10-20% silicon in the anode mixture resulting in anodes with overall capacity of about 700 to about 4,200 mAh/g.

[0013] Conventional cathode compositions have capacities of approximately 150-200 mAh/g and cannot be made thicker than 200 μm because conventional electrodes manufactured using the high speed roll-to-roll calendaring process tend to delaminate from the flat current collectors if they are made thicker than about 200 μm . Additionally, thicker electrodes have higher cell impedance, which reduces energy efficiency (e.g., as described in Yu et al “Effect of electrode parameters on LiFePO_4 cathodes”, *J. Electrochem. Soc. Vol. 153*, A835-A839 (2006)) Therefore, to match the high capacity anodes with the conventional cathodes, current state-of-the-art batteries have focused on reducing the thickness of the anode. For example, anodes having a thickness of about 40-50 μm and even thinner are being developed. Such thin coatings of these anode materials begin to approach the thickness level of a single graphite particle. The limitation of thickness and associated loading density in conventional coating processes has prevented development of batteries that take full advantage of the high capacity that is available in high energy anodes.

[0014] Semi-solid cathodes described herein can be made: (1) thicker (e.g., greater than 200 μm -up to 2,000 μm or even greater) due to the reduced tortuosity and higher electronic conductivity of the semi-solid electrode, (II) higher loadings of active materials, and (iii) with a simplified manufacturing process utilizing less equipment, thereby decreasing the volume, mass and cost contributions of inactive components with respect to active components. Examples of semi-solids cathodes having a thickness greater than 200 μm without an adverse affect on impedance, energy efficiency, energy density or capacity are described in U.S. Provisional Patent Application No. 61/787,382 filed Mar. 15, 2013, entitled “Semi-Solid Electrodes Having High Rate Capability,” the disclosure of which is hereby incorporated by reference. Examples of systems and methods that can be used for preparing the semi-solid compositions and/or electrodes are described in U.S. patent application Ser. No. 13/832,861 filed Mar. 15, 2013, entitled “Electrochemical Slurry Compositions and Methods for Preparing the Same,” the entire disclosure of which is hereby incorporated by reference.

[0015] Since the semi-solid cathodes can be much thicker than conventional electrodes (e.g., greater than 200 μm), and with higher loading densities, the semi-solid cathode can be more effectively paired with high energy anodes. Said another way, the thicker semi-solid cathode enables active material loading densities that are three, four, five, or more times higher than conventional coated electrodes. The thicker semi-solid cathodes with higher loading densities can be paired with anodes that utilize silicon, tin, or other high energy anode materials without having to maintain a thin anode and/or loading density. The high capacity anodes can be made with conventional coating methods and have higher loading density that approach the maximum active

material loading density of conventional electrodes that utilize binder and as a result, still be matched/balanced with the cathode. The pairing of the semi-solid cathode with the high capacity anode can result in cells with volumetric energy densities that are greater than 600 Wh/L or greater. In some embodiments, the anode can be a semi-solid anode made with high energy anode materials in a similar fashion as described with respect to the semi-solid cathode.

[0016] In some embodiments, the electrode materials described herein can be a flowable semi-solid or condensed liquid composition. A flowable semi-solid electrode can include a suspension of an electrochemically active material (anodic particles or particulates), and optionally an electronically conductive material (e.g., carbon) in a non-aqueous liquid electrolyte. Said another way, the active electrode particles and conductive particles are co-suspended in an electrolyte to produce a semi-solid electrode. The anode includes a high energy anode material e.g., silicon, tin, aluminum, titanium oxide or any other high capacity material, alloys or combination thereof. In some embodiments, the anode can be formed using a conventional coating/calendering process. In some embodiments, the anode can be a semi-solid anode that includes high energy anodic particles and carbon (e.g. graphite) particles co-suspended in an electrolyte to produce a semi-solid anode. Examples of battery architectures utilizing semi-solid suspensions are described in International Patent Publication No. WO 2012/024499, entitled “Stationary, Fluid Redox. Electrode,” and International Patent Publication No. WO 2012/088442, entitled “Semi-Solid Filled Battery and Method of Manufacture,” the entire disclosures of which are hereby incorporated by reference.

[0017] As used herein, the term “about” and “approximately” generally mean plus or minus 10% of the value stated, e.g., about 250 μm would include 225 μm to 275 μm , about 1,000 μm would include 900 μm to 1,100 μm .

[0018] As used herein, the term “semi-solid” refers to a material that is a mixture of liquid and solid phases, for example, such as particle suspension, colloidal suspension, emulsion, gel, or micelle.

[0019] As used herein, the term “condense ion-storing liquid” or “condensed liquid” refers to a liquid that is not merely a solvent, as in the case of an aqueous flow cell semi-solid cathode or anode, but rather, it is itself redox active. Of course, such a liquid form may also be diluted by or mixed with another, non-redox active liquid that is a diluent or solvent, including mixing with such a diluent to form a lower-melting liquid phase, emulsion or micelles including the ion-storing liquid.

[0020] As used herein, the terms “activated carbon network” and “networked carbon” relate to a general qualitative state of an electrode. For example, an electrode with an activated carbon network (or networked carbon) is such that the carbon particles within the electrode assume an individual particle morphology and arrangement with respect to each other that facilitates electrical contact and electrical conductivity between particles. Conversely, the terms “unactivated carbon network” and “unnetworked carbon” relate to an electrode wherein the carbon particles either exist as individual particle islands or multi-particle agglomerate islands that may not be sufficiently connected to provide adequate electrical conduction through the electrode.

[0021] As used herein, the term “volumetric energy density” refers to the amount of energy (e.g., Wh) stored in an electrochemical cell per unit volume (e.g., L) of the materials required for the electrochemical cell to operate such as, the electrodes, the separator, the electrolyte, and the current collectors. Specifically, the materials used for packaging the electrochemical cell are excluded from the calculation of volumetric energy density.

[0022] In some embodiments, an electrochemical cell for storing energy includes a positive electrode current collector, a negative electrode current collector and an ion-permeable membrane disposed between the positive electrode current collector and the negative electrode current collector. The ion-permeable membrane is spaced a first distance from the positive electrode current collector and at least partially defines a positive electroactive zone. The ion-permeable membrane is spaced a second distance from the negative electrode current collector and at least

partially defines a negative electroactive zone. The second distance is less than the first distance. A semi-solid cathode that includes a suspension of an active material and a conductive material in a non-aqueous liquid electrolyte is disposed in the positive electroactive zone, and an anode is disposed in the negative electroactive zone. In some embodiments, the anode can be a semi-solid anode that includes a suspension of a high capacity active material and a carbon material in a non-aqueous liquid electrolyte.

[0023] In some embodiments, the anode includes at least one high capacity anode material selected from silicon, bismuth, boron, gallium, indium, zinc, tin, antimony, aluminum, titanium oxide, molybdenum, germanium, manganese, niobium, vanadium, tantalum, iron, copper, gold, platinum, chromium, nickel, cobalt, zirconium, yttrium, molybdenum oxide, germanium oxide, silicon oxide, silicon carbide, any other high capacity materials or alloys thereof, and any combination thereof.

[0024] In some embodiments, the first distance is at least about 250 μm . In some embodiments, the first distance is about 300 μm , about 350 μm , about 400 μm , about 450 μm , about 500 μm , about 600 μm , about 700 μm , about 800 μm , about 900 μm , about 1,000 μm , about 1,500 μm , and up to about 2,000 μm , inclusive of all thicknesses therebetween.

[0025] In some embodiments, the first distance is in the range of about 250 μm to about 2,000 μm , about 300 μm to about 2,000 μm , about 350 μm to about 2,000 μm , 400 μm to about 2,000 μm , about 450 μm to about 2,000 μm , about 500 to about 2,000 μm , about 250 μm to about 1,500 μm , about 300 μm to about 1,500 μm , about 350 μm to about 1,500 μm , about 400 μm to about 1,500 μm , about 450 μm to about 1,500 μm , about 500 to about 1,500 μm , about 250 μm to about 1,000 μm , about 300 μm to about 1,000 μm , about 350 μm to about 1,000 μm , about 400 μm to about 1,000 μm , about 450 μm to about 1,000 μm , about 500 μm to about 1,000 μm , about 250 μm to about 750 μm , about 300 μm to about 750 μm , about 350 μm to about 750 μm , 400 μm to about 750 μm , about 450 μm to about 750 μm , about 500 μm to about 750 μm , about 250 μm to about 700 μm , about 300 μm to about 700 μm , about 350 μm to about 700 μm , about 400 μm to about 700 μm , about 450 μm to about 700 μm , about 500 μm to about 700 μm , about 250 μm to about 650 μm , about 300 μm to about 650 μm , about 350 μm to about 650 μm , about 400 μm to about 650 μm , about 450 μm to about 650 μm , about 500 μm to about 650 μm , about 250 μm to about 600 μm , about 300 μm to about 600 μm , about 350 μm to about 600 μm , about 400 μm to about 600 μm , about 450 μm to about 600 μm , about 500 μm to about 600 μm , about 250 μm to about 550 μm , about 300 μm to about 550 μm , about 350 μm to about 550 μm , about 400 μm to about 550 μm , about 450 μm to about 550 μm , or about 500 μm to about 550 μm , inclusive of all ranges or any other distance therebetween.

[0026] In some embodiments, the second distance is in the range of about 30 μm to about 200 μm , about 40 μm to about 200 μm , about 50 μm to about 200 μm , about 60 μm to about 200 μm , about 70 μm to about 200 μm , about 100 μm to about 200 μm , about 150 μm to about 200 μm , about 200 μm to about 300 μm , about 200 μm to about 400 μm , about 200 μm to about 500 μm , inclusive of all ranges or any other distance therebetween.

[0027] In some embodiments, the anode can include about 66 wt %-70 wt % Si, about 15 wt %-22 wt % Co, and about 4 wt %-12 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 15 wt %-20 wt % Ni and about 10 wt %-15 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 15 wt % Fe and about 15 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 20 wt % Ti, and about 10 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 15 wt % Mo and about 15 wt % C. In some embodiments, the anode can include about 70 wt % Si, 15 wt % Co, 5 wt % Ni and about 10 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 10 wt % Co, about 10 wt % Ni and about 10 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 5 wt % Co, about 15 wt % Ni and about 10 wt % C. In some embodiments, the anode can include about 70 wt % Si, about 5 wt % Fe, about 10 wt % Ni and about 15 wt % C. In some embodiments, the anode can include about 70 wt % Si, 10 wt % Co about 5 wt % Ni. In some

embodiments, the anode can include about 74 wt % Si, 2 wt % Sn and about 24 wt % Co. In some embodiments, the anode can include about 73 wt % Si, about 2 wt % Sn and about 25 wt % Ni. In some embodiments, the anode can include about 70 wt % Si, 10 wt % Fe, about 10 wt % Ti and about 10 wt % Co. In some embodiments, the anode can include about 70 wt % Si, about 15 wt % Fe, about 5 wt % Ti and about 10 wt % C. In some embodiments, the anode can include about 74.67 wt % Si, 16 wt % Fe, 5.33 wt % Ti and 4 wt % C. In some embodiments, the anode can include about 55 wt % Si, 29.3 wt % Al and about 15.7 wt % Fe. In some embodiments, the anode can include about 70 wt % Si, about 20 wt % C from a precursor and about 10 wt % graphite by weight. In some embodiments, the anode can include about 55 wt % Si, about 29.3 wt % Al and about 15.7 wt % Fe. In some embodiments, the anode can include about 60-62 wt % Si, about 16-20 wt % Al, about 12-14 wt % Fe, and about 8% Ti. In some embodiments, the anode can include about 50 wt % Sn, about 27.3 wt %-35.1 wt % Co, about 5 wt %-15 wt % Ti, and about 7.7 wt %-9.9 wt % C. In some embodiments, the anode can include about 50 wt % Sn, about 39-42.3 wt % Co, and about 7.7-11 wt % C. In some embodiments, the anode can include about 35-70 mole % Si, about 1-45 mole % Al, about 5-25 mole % transition metal, about 1-15 mole % Sn, about 2-15 mole % yttrium, a lanthanide element, an actinide element or a combination thereof.

[0028] In some embodiments, the anode can include a tin metal alloy such as, for example, a Sn—Co—C, a Sn—Fe—C, a Sn—Mg—C, or a La—Ni—Sn alloy. In some embodiments, the anode can include an amorphous oxide such as, for example, SnO or SiO amorphous oxide. In some embodiments, the anode can include a glassy anode such as, for example, a Sn—Si—Al—B—O, a Sn—Sb—S—O, a $\text{SnO}_{0.2}\text{—P}_{0.2}\text{O}_{0.5}$, or a $\text{SnO—B}_{0.2}\text{O}_{0.3}\text{—P}_{0.2}\text{O}_{0.5}\text{—Al}_{0.2}\text{O}_{0.3}$ anode. In some embodiments, the anode can include a metal oxide such as, for example, a CoO, a $\text{SnO}_{0.2}$, or a $\text{V}_{0.2}\text{O}_{0.5}$. In some embodiments, the anode can include a metal nitride such as, for example, $\text{Li}_{0.3}\text{N}$ or $\text{Li}_{0.2}\text{Math.6CoO.Math.4N}$.

[0029] In some embodiments, the anode can be single layer anode that includes all components of the anode (e.g., active material, conductive material, electrolyte, and high capacity layer) mixed together in the layer. In some embodiments, the anode can include a multi-layer structure, for example, a multiple layers of high capacity anode materials combined with carbon materials.

[0030] In some embodiments, an electrochemical cell includes a semi-solid cathode including a suspension of an active material and a conductive material in a non-aqueous liquid electrolyte the semi-solid cathode having a capacity of about 150-200 mAh/g and a thickness in the range of about 250 μm to about 2,000 μm , and an anode that has a capacity in the range of about 700-1,200 mAh/g and a thickness in the range of about 30 μm to about 600 μm . The anode and semi-solid cathode are separated by a separator disposed therebetween.

[0031] In some embodiments, the anode includes an anode active material selected from lithium metal, carbon, lithium-intercalated carbon, lithium nitrides, lithium alloys and lithium alloy forming compounds of silicon, bismuth, boron, gallium, indium, zine, tin, antimony; aluminum, titanium oxide, molybdenum, germanium, manganese, niobium, vanadium, tantalum, gold, platinum, iron, copper, chromium, nickel, cobalt, zirconium, yttrium, molybdenum oxide, germanium oxide, silicon oxide, silicon carbide, any other high capacity materials or alloys thereof, and any other combination thereof. In some embodiments, the anode active material can include silicon and/or alloys thereof. In some embodiments, anode active material can include tin and/or alloys thereof

[0032] In some embodiments, the thickness of the cathode can be in the range of about 2 times to about 6 times the thickness of the anode. In some embodiments, the thickness of the cathode can be in the range of about 2 to about 5, about 2 to about 4, about 2 to about 3, about 3 to about 6, about 4 to about 6, or about 5 to about 6 times the thickness of the anode.

[0033] In some embodiments, a semi-solid cathode can include about 20% to about 75% by volume of an active material. In some embodiments, a semi-solid cathode can include about 40% to about 75% by volume, or 60% to about 75% by volume of an active material.

[0034] In some embodiments, a semi-solid cathode can include about 0.5% to about 25% by volume of a conductive material. In some embodiments, a semi-solid cathode can include about 1.0% to about 6% by volume of a conductive material.

[0035] In some embodiments, a semi-solid cathode can include about 25% to about 70% by volume of an electrolyte. In some embodiments, a semi-solid cathode can include about 30% to about 50%, or about 20% to about 40% by volume of an electrolyte.

[0036] In some embodiments, the anode is a solid anode formed on a negative electrode current collector. In some embodiments, the anode is a semi-solid anode that includes about 50% to about 75% by volume of a carbonaceous active material, about 1% to about 10% by volume of a high energy capacity anode material, about 0.5% to about 2% by volume of a conductive material, and about 20% to about 40% by volume of a non-aqueous liquid electrolyte. In some embodiments, the high capacity anode material includes at least one high capacity anode material selected from silicon, silicon alloys, tin, tin alloys, aluminum and titanium oxide.

[0037] In some embodiments, a semi-solid anode can include about 0% to about 75% by volume of an active material. In some embodiments, a semi-solid anode can include about 40% to about 75% by volume, or about 50% to about 75% by volume of an active material.

[0038] In some embodiments, a semi-solid anode can include about 0% to about 75% by volume of a high energy capacity material. In some embodiments, a semi-solid anode can include about 1% to about 30% by volume, about 1% to about 20% by volume, about 1% to about 10% by volume, or about 1% to about 5% of a high energy capacity material.

[0039] In some embodiments, a semi-solid anode can include about 0% to about 10% by volume of a conductive material. In some embodiments, a semi-solid anode can include about 1% to about 6%, or about 0.5% to about 2% by volume of a conductive material.

[0040] In some embodiments, a semi-solid anode can include about 10% to about 70% by volume of an electrolyte. In some embodiments, a semi-solid anode can include about 30% to about 50%, or about 20% to about 40% by volume of an electrolyte.

[0041] In some embodiments, an electrochemical cell including a semi-solid cathode and a high energy anode can have a volumetric energy density that is greater than about 600 Wh/L, greater than about 700 Wh/L, greater than about 800 Wh/L, greater than about 900 Wh/L, greater than about 1,000 Wh/L, greater than about 1,100 Wh/L, and up to about 1,200 Wh/L.

[0042] In some embodiments, an electrochemical cell can include an anode, a semi-solid cathode material that includes about 60% to about 80% by weight of an active material, about 1% to about 68% by weight of a conductive material, and about 20% to about 40% by weight of a non-aqueous liquid electrolyte. A separator is disposed between the anode and the semi-solid cathode such that the thickness of the cathode is at least about two times the thickness of the anode. In some embodiments, the thickness of the cathode is at least about three times the thickness of the anode. In some embodiments, the thickness of the cathode is at least about four times the thickness of the anode. In some embodiments, the thickness of the cathode is at least about five times the thickness of the anode.

[0043] In some embodiments, an electrochemical cell includes a semi-solid cathode that can include about 60% to about 80% by weight of an active material, about 1% to about 6% by weight of a conductive material, and about 20% to about 40% by weight of a non-aqueous liquid electrolyte. The semi solid cathode can have a thickness in the range of about 250 μm to about 2,000 μm . The electrochemical cell also includes an anode that can include a high charge capacity material and can have an overall capacity in the range of about 700 mAh/g to about 1,200 mAh/g. The anode can have a thickness in the range of about 20% to about 25% of the cathode thickness. The anode and semi-solid cathode are separated by a separator disposed therebetween.

[0044] In some embodiments, a method of manufacturing an electrochemical cell includes transferring an anode semi-solid suspension to a negative electroactive zone defined at least in part by an anode current collector and an ion-permeable membrane spaced apart from the anode

collector. The method also includes transferring a cathode semi-solid suspension to a positive electroactive zone defined at least in part by a cathode current collector and an ion-permeable membrane spaced apart from the cathode current collector. The transferring of the anode semi-solid suspension to the anode compartment and the cathode semi-solid suspension to the cathode compartment can be performed such that a difference between a minimum distance and a maximum distance between the anode current collector and the ion permeable membrane is maintained within a predetermined tolerance. Furthermore, the thickness of positive electroactive zone is substantially greater than the thickness of the negative electroactive zone. In some embodiments, the positive electroactive zone can be about two fold, three fold, four fold or five fold thicker than the negative electroactive zone. The method further includes sealing the negative electroactive zone and the positive electroactive zone.

[0045] In some embodiments, a method of manufacturing an electrochemical cell includes disposing an inject nozzle in an electrode that is defined, at least in part by a current collector and an ion-permeable membrane. A semi-solid cathode material is transferred to the electroactive zone through the injection nozzle. The injection nozzle is then withdrawn from the electrode compartment during at least a portion of the transferring and then the electrode compartment is sealed.

[0046] FIG. 1 shows a schematic illustration of an electrochemical cell **100** for storing energy with high capacity. The electrochemical cell **100** includes a positive current collector **110**, a negative current collector **120** and a separator **130** disposed between the positive current collector **110** and the negative current collector **120**. The positive current collector **110** is spaced a first distance $t_{sub.1}$ from the separator **130** and at least partially defines a positive electroactive zone. The negative current collector **120** is spaced a second distance $t_{sub.2}$ from the separator **130** and at least partially defines a negative electroactive zone. The first distance $t_{sub.1}$ is substantially greater than $t_{sub.2}$. A semi-solid cathode **140** is disposed in the positive electroactive zone and an anode **150** is disposed in the negative electroactive zone, such that the first distance $t_{sub.1}$ defines thickness of the cathode **140** and the second distance $t_{sub.2}$ defines the thickness of the anode **150**.

[0047] The positive current collector **110** and the negative current collector **120** can be any current collectors that are electronically conductive and are electrochemically inactive under the operation conditions of the cell. Typical current collectors for lithium cells include copper, aluminum, or titanium for the negative current collector and aluminum for the positive current collector, in the form of sheets or mesh, or any combination thereof.

[0048] Current collector materials can be selected to be stable at the operating potentials of the positive and negative electrodes of the electrochemical cell **100**. For example, in non-aqueous lithium systems, the positive current collector can include aluminum, or aluminum coated with conductive material that does not electrochemically dissolve at operating potentials of 2.5-5.0V with respect to Li/Li^{sup.}+, Such materials include platinum, gold, nickel, conductive metal oxides such as vanadium oxide, and carbon, The negative current collector can include copper or other metals that do not form alloys or intermetallic compounds with lithium, carbon, and/or coatings comprising such materials disposed on another conductor.

[0049] The separator **130** can be any suitable separator that acts as an ion-permeable membrane, i.e. allows exchange of ions while maintaining physical separation of the cathode **140** and anode **150** compositions. For example, the separator **130** can be any conventional membrane that is capable of ion transport. In some embodiments, the separator **130** is a liquid impermeable membrane that permits the transport of ions therethrough, namely a solid or gel ionic conductor. In some embodiments the separator **130** is a porous polymer membrane infused with a liquid electrolyte that allows for the shuttling of ions between the cathode **140** and anode **150** electroactive materials, while preventing the transfer of electrons. In some embodiments, the separator **130** is a microporous membrane that prevents particles forming the positive and negative electrode compositions from crossing the membrane. For example, the membrane materials can be

selected from polyethyleneoxide (PEO) polymer in which a lithium salt is complexed to provide lithium conductivity, or Nation™ membranes which are proton conductors. For example, PEO based electrolytes can be used as the membrane, which is pinhole-free and a solid ionic conductor, optionally stabilized with other membranes such as glass fiber separators as supporting layers. PEO can also be used as a slurry stabilizer, dispersant, etc. in the positive or negative redox compositions. PEO is stable in contact with typical alkyl carbonate-based electrolytes. This can be especially useful in phosphate-based cell chemistries with cell potential at the positive electrode that is less than about 3.6 V with respect to Li metal. The operating temperature of the redox cell can be elevated as necessary to improve the ionic conductivity of the membrane.

[0050] The cathode **140** can be a semi-solid stationary cathode or a semi-solid flowable cathode, for example of the type used in redox cells. The cathode **140** can include an active material such as a lithium bearing compound as described in further detail below. The cathode **140** can also include a conductive material such as, for example, graphite, carbon powder, pyrolytic carbon, carbon black, carbon fibers, carbon microfibers, carbon nanotubes (CNTs), single walled CNTs, multi walled CNTs, fullerene carbons including “bucky balls,” graphene sheets and/or aggregate of graphene sheets, any other conductive material, alloys or combination thereof. The cathode **140** can also include a non-aqueous liquid electrolyte as described in further detail below.

[0051] In some embodiments, the semi-solid cathode **140** can include about 20%-75% by volume of an active material, about 0.5%-25% by volume of a conductive material and about 25%-70% by volume of an electrolyte. In some embodiments, the semi-solid cathode **140** can include about 60% to about 80% by weight of an active material, about 1% to about 6% by weight of a conductive material, and about 20% to about 40% by weight of a non-aqueous liquid electrolyte.

[0052] In some embodiments, the thickness $t_{\text{sub.1}}$ of the cathode **140** can be in the range of about two times to about six times the thickness $t_{\text{sub.2}}$ of the anode **150**. In some embodiments, the thickness $t_{\text{sub.1}}$ of the cathode **140** can be about two times the thickness $t_{\text{sub.2}}$ of the anode **150**. In some embodiments, the thickness $t_{\text{sub.1}}$ of the cathode **140** can be about three times the thickness $t_{\text{sub.2}}$ of the anode **150**. In some embodiments, the thickness $t_{\text{sub.1}}$ of the cathode **140** can be about four times the thickness $t_{\text{sub.2}}$ of the anode **150**. In some embodiments, the thickness $t_{\text{sub.1}}$ of the cathode **140** can be about five times the thickness $t_{\text{sub.2}}$ of the anode **150**. In some embodiments, the thickness $t_{\text{sub.1}}$ of the cathode **140** can be about six times the thickness $t_{\text{sub.2}}$ of the anode **150**.

[0053] The anode **150** can be a conventional anode, e.g., formed through a conventional coating and calendaring process. In some embodiments, the anode **150** can be a semi-solid stationary anode. In some embodiments, the anode **150** can be a semi-solid flowable anode, e.g., of the type used in re lox low cells. The anode **150** is a high capacity anode that includes high capacity active materials, e.g., silicon, bismuth, boron, gallium, indium, zinc, tin, antimony, aluminum, titanium oxide, molybdenum, germanium, manganese, niobium, vanadium, tantalum, iron, copper, chromium, nickel, cobalt, zirconium, yttrium, molybdenum oxide, germanium oxide, silicon oxide, silicon carbide, any other high capacity materials or alloys thereof, or any other combination thereof.

[0054] The anode **150** can also include a carbonaceous material such as, e.g., graphite, carbon powder, pyrolytic carbon, carbon black, carbon fibers, carbon microfibers, carbon nanotubes (CNTs), single walled CNTs, multi walled CNTs, fullerene carbons including “bucky balls”, graphene sheets and/or aggregate of graphene sheets, any other carbonaceous material or combination thereof. In some embodiments, the anode **150** can also include a non-aqueous liquid electrolyte as described in further detail below.

[0055] In some embodiments, the thickness $t_{\text{sub.2}}$ of anode **150** can be about 20-25% of the thickness $t_{\text{sub.1}}$ of the cathode **140**. In some embodiments, the thickness $t_{\text{sub.2}}$ of the anode **150** can be in the range of about 30 μm to about 600 μm , about 40 μm to about 600 μm , about 50 μm to about 600 μm , about 70 μm to about 600 μm , about 100 μm to about 600 μm , about 150 μm to

about 600 μm , about 200 μm to about 600 μm , about 50 μm to about 500 μm , about 100 μm to about 500 μm , about 150 μm to about 500 μm , about 200 μm to about 500 μm , about 100 μm to about 400 μm , about 150 μm to about 400 μm , about 200 μm to about 400 μm , about 100 μm to about 300 μm , about 150 μm to about 300 μm , about 200 μm to about 300 μm , about 30 μm to about 200 μm , about 40 μm to about 200 μm , about 50 μm to about 200 μm , about 60 μm to about 200 μm , about 70 μm to about 200 μm , about 100 μm to about 200 μm , about 150 μm to about 200 μm , about 200 μm , about 300 μm , inclusive of any other thickness or thickness range therebetween.

[0056] In some embodiments, the cathode **140** and the anode **150** can both include semi-solid suspensions. In such embodiments, the cathode **140** and the anode **150** can include active materials and optionally conductive materials in particulate form suspended in a non-aqueous liquid electrolyte. In some embodiments, the cathode **140** and/or anode **150** particles have an effective diameter of at least about 1 μm . In some embodiments, the cathode **140** and/or anode **150** particles have an effective diameter between about 1 μm and about 10 μm . In other embodiments, the cathode **140** and/or anode **150** particles have an effective diameter of at least about 10 μm or more.

[0057] In some embodiments, a redox mediator is used to improve the charge transfer within the semi-solid suspension. In some embodiments, the redox mediator is based on $\text{Fe}^{\text{sup.2+}}$ or $\text{V}^{\text{sup.2+}}$, $\text{V}^{\text{sup.3+}}$, or $\text{V}^{\text{sup.4+}}$. In some embodiments, the redox mediator is ferrocene.

[0058] In some embodiments, the conductive particles have shapes, which may include spheres, platelets, or rods to optimize solids packing fraction, increase the semi-solid's net electronic conductivity, and improve rheological behavior of semi-solids. Low aspect or substantially equiaxed particles tend to flow well, however, they tend to have a low packing density.

[0059] In some embodiments, the particles have a plurality of sizes so as to increase packing fraction by placing smaller particles in the interstices of the larger particles. In particular, the particle size distribution can be bimodal, in which the average particle size of the larger particle mode is at least 5 times larger than average particle size of the smaller particle mode. The mixture of large and small particles improves flow of the material during cell loading and increases solid volume fraction and packing density in the loaded cell.

[0060] In some embodiments, the nature of cathode **140** and/or anode **150** semi-solid suspension can be modified prior to and subsequent to filling of the negative electroactive zone and the positive electroactive zone to facilitate flow during loading and packing density in the loaded cell.

[0061] In some embodiments, the particle suspension is initially stabilized by repulsive interparticle steric forces that arise from surfactant molecules. After the particle suspension is loaded into the positive electroactive zone and/or the negative electroactive zone, chemical or heat treatments can cause these surface molecules to collapse or evaporate and promote densification. In some embodiments, the suspension's steric forces are modified intermittently during loading.

[0062] For example, the particle suspension can be initially stabilized by repulsive interparticle electrostatic double layer forces to decrease viscosity. The repulsive force reduces interparticle attraction and reduces agglomeration. After the particle suspension is loaded into the positive electroactive zone and/or negative electroactive zone, the surface of the particles can be further modified to reduce interparticle repulsive forces and thereby promote particle attraction and packing. For example, ionic solutions such as salt solutions can be added to the suspension to reduce the repulsive forces and promote aggregation and densification so as to produce increased solids fraction loading after filling of the electroactive zones. In some embodiments, salt is added intermittently during suspension loading to increase density in incremental layers.

[0063] In some embodiments, the positive and/or negative electroactive zones are loaded with a particle suspension that is stabilized by repulsive forces between particles induced by an electrostatic double layer or short-range steric forces due to added surfactants or dispersants. Following loading, the particle suspension is aggregated and densified by increasing the salt concentration of the suspension. In some embodiments, the salt that is added to is a salt of a working ion for the battery (e.g., a lithium salt for a lithium ion battery) and upon being added,

causes the liquid phase to become an ion-conducting electrolyte (e.g. for a lithium rechargeable battery, may be one or more alkyl carbonates, or one or more ionic liquids). Upon increasing the salt concentration, the electrical double layer causing repulsion between the particles is “collapsed”, and attractive interactions cause the particle to floc, aggregate, consolidate, or otherwise densify. This allows the electrode of the battery to be formed from the suspension while it has a low viscosity, for example, by pouring, injection, or pumping into the positive a d/or negative electroactive zones that can form a net-shaped electrode, and then allows particles within the suspension to be consolidated for improved electrical conduction, higher packing density and longer shelf life.

[0064] In some embodiments, the flowable cathode **140** and/or anode **150** semi-solid suspension is caused to become non-flowable by “fixing”. In some embodiments, fixing can be performed by the action of photopolymerization. In some embodiments, fixing is performed by action of electromagnetic radiation with wavelengths that are transmitted by the unfilled positive and/or negative electroactive zones of the electrochemical cell **100**. In some embodiments, one or more additives are added to the semi-solid suspensions to facilitate fixing.

[0065] In some embodiments, the injectable and flowable cathode **140** and/or anode **150** semi-solid is caused to become non-flowable by “plasticizing”. In some embodiments, the rheological properties of the injectable and flowable semi-solid suspension are modified by the addition of a thinner, a thickener, and/or a plasticizing agent. In some embodiments, these agents promote processability and help retain compositional uniformity of the semi-solid under flowing conditions and positive and negative electroactive zone filling operations. In some embodiments, one or more additives are added to the flowable semi-solid suspension to adjust its flow properties to accommodate processing requirements.

Semi-Solid Composition

[0066] In some embodiments, the cathode **140** and in some embodiments, the anode **150** semi-solids provide a means to produce a substance that functions collectively as an ion-storage/ion-source, electron conductor, and ionic conductor in a single medium that acts as a working electrode.

[0067] The cathode **140** and/or anode **150** semi-solid ion-storing redox composition as described herein can have, when taken in moles per liter (molarity), at least 10M concentration of redox species. In some embodiments, the cathode **140** and/or the anode **150** semi-solids ion-storing redox composition can have at least 12M, at least 15M, or at least 20M. The electrochemically active material can be an ion storage material and or any other compound or ion complex that is capable of undergoing Faradaic reaction in order to store energy. The electroactive material can also be a multiphase material including the above described redox-active solid mixed with a non-redox-active phase, including solid-liquid. suspensions, or liquid-liquid multiphase mixtures, including micelles or emulsions having a liquid ion-storage material intimately mixed with a supporting liquid phase. Systems that utilize various working ions can include aqueous systems in which Li.sup.+, Na.sup.+, or other alkali ions are the working ions, even alkaline earth working ions such as Ca.sup.2+, Mg.sup.2+, or Al.sup.3+. In each of these instances, a negative electrode storage material and a positive electrode storage material may be required, the negative electrode storing the working ion of interest at a lower absolute electrical potential than the positive electrode. The cell voltage can be determined approximately by the difference in ion-storage potentials of the two ion-storage electrode materials.

[0068] Systems employing both negative and positive ion-storage materials are particularly advantageous because there are no additional electrochemical byproducts in the cell. Both the positive and negative electrode materials are insoluble in the electrolyte and the electrolyte does not become contaminated with electrochemical composition products, In addition, systems employing both negative and positive lithium ion-storage materials are particularly advantageous when using non-aqueous electrochemical compositions.

[0069] In some embodiments, the semi-solid ion-storing redox compositions include materials proven to work in conventional, solid lithium-ion batteries. In some embodiments, the positive semi-solid electroactive material contains lithium positive electroactive materials and the lithium cations are shuttled between the negative electrode and positive electrode, intercalating into solid, host particles suspended in a liquid electrolyte.

[0070] In some embodiments, at least one of the semi-solid cathode **140** and/or anode **150** includes a condenser ion-storing liquid of a redox-active compound, which may be organic or inorganic, and includes but is not limited to lithium metal, sodium metal, lithium-metal alloys, gallium and indium alloys with or without dissolved lithium, molten transition metal chlorides, thionyl chloride, and the like, or redox polymers and organics that can be liquid under the operating conditions of the battery. Such a liquid form may also be diluted by or mixed with another, non-redox-active liquid that is a diluent or solvent, including mixing with such diluents to form a lower-melting liquid phase. In some embodiments, the redox-active component can comprise, by mass, at least 10% of the total mass of the electrolyte. In other embodiments, the redox-active component will comprise, by mass, between approximately 10% and 25% of the total mass of the electrolyte. In some embodiments, the redox-active component will comprise by mass, at least 25% or more of the total mass of the electrolyte.

[0071] In some embodiments, the redox-active electrode material, whether used as a semi-solid or a condensed liquid format as defined above, comprises an organic redox compound that stores the working ion of interest at a potential useful for either the positive or negative electrode of a battery. Such organic redox-active storage materials include “p”-doped conductive polymers such as polyaniline or polyacetylene based materials, polynitroxide or organic radical electrodes (such as those described in: H. Nishide et al., *Electrochim. Acta*, 50, 827-831, (2004), and K. Nakahara, et al., *Chem. Phys. Lett.*, 359, 351-354 (2002)), carbonyl based organics, and oxocarbons and carboxylate, including compounds such as $\text{Li}_{0.2}\text{C}_{60}$, $\text{Li}_{0.2}\text{C}_{84}\text{H}_{40}$, and $\text{Li}_{0.2}\text{C}_{60}\text{H}_{40}$ (see for example M. Armand et al., *Nature Materials*, DOI: 10.1038/nmat2372) and organosulfur compounds.

[0072] In some embodiments, organic redox compounds that are electronically insulating are used. In some instance, the redox compounds are in a condensed liquid phase such as liquid or flowable polymers that are electronically insulating. In such cases, the redox active slurry may or may not contain an additional carrier liquid. Additives can be combined with the condensed phase liquid redox compound to increase electronic conductivity. In some embodiments, such electronically insulating organic redox compounds are rendered electrochemically active by mixing or blending with particulates of an electronically conductive material, such as solid inorganic conductive materials including but not limited to metals, metal carbides, metal nitrides, metal oxides, and allotropes of carbon including carbon black, graphitic carbon, carbon fibers, carbon microfibers, vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multiwall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene sheets or aggregates of graphene sheets, and materials comprising fullerenic fragments.

[0073] In some embodiments, such electronically insulating organic redox compounds are rendered electronically active by mixing or blending with an electronically conductive polymer, including but not limited to polyaniline or polyacetylene based conductive polymers or poly(3,4-ethylenedioxythiophene) (PEDOT), polypyrrole, polythiophene, poly(p-phenylene), poly(triphenylene), polyazulene, polyfluorene, polynaphthalene, polyanthracene, polyfuran, polycarbazole, tetrathiafulvalene-substituted polystyrene, ferrocene-substituted polyethylene, carbazole-substituted polyethylene, polyoxyphenazine, polyacenes, or poly(heteroacenes). The conductive additives form an electrically conducting framework within the insulating liquid redox compounds that significantly increases the electrically conductivity of the composition. In some embodiments, the conductive addition forms a percolative pathway to the current collector. In some embodiments the redox-active electrode material comprises a sol or gel, including for example

metal oxide sols or gels produced by the hydrolysis of metal alkoxides, amongst other methods generally known as “sol-gel processing.” Vanadium oxide gels of composition $V_{x/2}O_y$ are amongst such redox-active sol-gel materials.

[0074] Other suitable positive active materials for use in the cathode **140** include solid compounds known to those skilled in the art as those used in NiMH (Nickel-Metal Hydride) or Nickel Cadmium (NiCd) batteries. Still other positive electrode compounds for Li storage include those used in carbon monofluoride batteries, generally referred to as CF_x, or metal fluoride compounds having approximate stoichiometry MF₂ or MF₃ where M comprises, for example, Fe, Bi, Ni, Co, Ti, or V. Examples include those described in H. Li, P. Balaya, and J. Maier, Li-Storage via Heterogeneous Reaction in Selected Binary Metal Fluorides and Oxides, *Journal of The Electrochemical Society*, 151 [11]A1878-A1885 (2004), M. Bervas, A. N. Mansour, W.-S. Woon, J. F. Al-Sharab, F. Badway, F. Cosandey, L. C. Klein, and G. G. Amatucci, “Investigation of the Lithiation and Delithiation Conversion Mechanisms in a Bismuth Fluoride Nanocomposites”, *J. Electrochem. Soc.*, 153, A799 (2006), and I. Plitz, F. Badway, J. Al-Sharab, A. DuPasquier, F. Cosandey and G. G. Amatucci, “Structure and Electrochemistry of Carbon-Metal Fluoride Nanocomposites Fabricated by a Solid State Redox Conversion Reaction”, *J. Electrochem. Soc.*, 152, A307 (2005).

[0075] As another example, fullerenic carbon including single-wall carbon nanotubes (SWNTs), multiwall carbon nanotubes (MWNTs), or metal or metalloid nanowires may be used as ion-storage materials. One example is the silicon nanowires used as a high energy density storage material in a report by C. K. Chan, H. Peng, G. Lin, K. McIlwrath, X. F. Zhang, R. A. Huggins, and Y. Cui, High-performance lithium battery anodes using silicon nanowires, *Nature Nanotechnology*, published online 16 Dec. 2007; doi: 10.1038/nnano.2007.411. In some embodiments, electroactive materials for the cathode **140** in a lithium system can include the general family of ordered rocksalt compounds LiMO₂ including those having the α -NaFeO₂ (so-called “layered compounds”) or orthorhombic-LiMnO₂ structure type or their derivatives of different crystal symmetry, atomic ordering, or partial substitution for the metals or oxygen. M comprises at least one first-row transition metal but may include non-transition metals including but not limited to Al, Ca, Mg, or Zr. Examples of such compounds include LiCoO₂, LiCoO₂ doped with Mg, LiNiO₂, Li(Ni, Co, Al)O₂ (known as “NCA”) and Li(Ni, Mn, Co)O₂ (known as “NMC”), Other families of exemplary cathode **140** electroactive materials includes those of spinel structure, such as LiMn₂O₄ and its derivatives, so-called “layered-spinel nanocomposites” in which the structure includes nanoscopic regions having ordered rocksalt and spinel ordering, olivines LiMPO₄ and their derivatives, in which M comprises one or more of Mn, Fe, Co, or Ni, partially fluorinated compounds such as LiVPO₄F₂, other “polyanion” compounds as described below, and vanadium oxides V_{x/2}O_y including V₂O₅ and V₆O₁₁.

[0076] In some embodiments, the cathode **140** electroactive material comprises a transition metal polyanion compound, for example as described in U.S. Pat. No. 7,338,734. In some embodiments the active material comprises an alkali metal transition metal oxide or phosphate, and for example, the compound has a composition A_x(M'_{1-a}M''_a)_y(XD₄)_z, A_x(M'_{1-a}M''_a)_y(DXD₄)_z, or A_x(M'_{1-a}M''_a)_y(X₂D₇)_z, and have values such that x, plus y(1-a) times a formal valence or valences of M', plus ya times a formal valence or valence of M'', is equal to z times a formal valence of the XD₄, X₂D₇, or DXD₄ group; or a compound comprising a composition (A_{1-a}M''_a)_xM'_y(XD₄)_z, (A_{1-a}M''_a)_xM'_y(DXD₄)_z, (A_{1-a}M''_a)_xM'_y(X₂D₇)_z and have values such that (1-a)x plus the quantity ax times the formal valence or valences of M'' plus y times the formal valence or valences of M' is equal to z times the formal valence of the XD₄, X₂D₇ or DXD₄ group. In the compound,

A is at least one of an alkali metal and hydrogen, M' is a first-row transition metal, X is at least one of phosphorus, sulfur, arsenic, molybdenum, and tungsten, M'' any of a Group IIA, IIIA, IVA, VA, VIA, VIIA, VIIIA, IB, IIB, IIIB, IVB, VB, and VIB metal, D is at least one of oxygen, nitrogen, carbon, or a halogen. The positive electroactive material can be an olivine structure compound $\text{LiMPO}_{0.4}$, where M is one or more of V, Cr, Mn, Fe, Co, and Ni, in which the compound is optionally doped at the Li, M or O-sites. Deficiencies at the Li-site are compensated by the addition of a metal or metalloid, and deficiencies at the O-site are compensated by the addition of a halogen. In some embodiments, the positive active material comprises a thermally stable, transition-metal-doped lithium transition metal phosphate having the olivine structure and having the formula $(\text{Li}_{1-x}\text{Z}_x)\text{MPO}_{0.4}$, where M is one or more of V, Cr, Mn, Fe, Co, and Ni, and Z is a non-alkali metal dopant such as one or more of Ti, Zr, Nb, Al, or Mg, and x ranges from 0.005 to 0.05.

[0077] In other embodiments, the lithium transition metal phosphate material has an overall composition of $\text{Li}_{1-x-z}\text{M}_{1-z}\text{PO}_{0.4}$, where M comprises at least one first row transition metal selected from the group consisting of Ti, V, Cr, Mn, Fe, Co and Ni, where x is from 0 to 1 and z can be positive or negative. M includes Fe, z is between about 0.15-0.15. The material can exhibit a solid solution over a composition range of $0 < x < 0.15$, or the material can exhibit a stable solid solution over a composition range of x between 0 and at least about 0.05, or the material can exhibit a stable solid solution over a composition range of x between 0 and at least about 0.07 at room temperature (22-25° C.). The material may also exhibit a solid solution in the lithium-poor regime, e.g., where $x \geq 0.8$, or $x \geq 0.9$, or $x \geq 0.95$.

[0078] In some embodiments the redox-active electrode material comprises a metal salt that stores an alkali ion by undergoing a displacement or conversion reaction. Examples of such compounds include metal oxides such as CoO, $\text{Co}_{0.3}\text{O}_{0.4}$, NiO, CuO, MnO, typically used as a negative electrode in a lithium battery, which upon reaction with Li undergo a displacement or conversion reaction to form a mixture of Li_2O and the metal constituent in the form of a more reduced oxide or the metallic form. Other examples include metal fluorides such as CuF_2 , FeF_2 , FeF_3 , BiF_3 , CoF_2 , and NiF_2 , which undergo a displacement or conversion reaction to form LiF and the reduced metal constituent. Such fluorides may be used as the positive electrode in a lithium battery. In other embodiments the redox-active electrode material comprises carbon monofluoride or its derivatives. In some embodiments the material undergoing displacement or conversion reaction is in the form of particulates having on average dimensions of 100 nanometers or less. In some embodiments the material undergoing displacement or conversion reaction comprises a nanocomposite of the active material mixed with an inactive host, including but not limited to conductive and relatively ductile compounds such as carbon, or a metal, or a metal sulfide. FeS_2 and FeF_3 can also be used as cheap and electronically conductive active materials in a nonaqueous or aqueous lithium system.

[0079] In some embodiments, the working ion is selected from the group consisting of Li^+ , Na^+ , H^+ , Mg^{2+} , Al^{3+} , or Ca^{2+} .

[0080] In some embodiments, the working ion is selected from the group consisting of Li^+ or Na^+ .

[0081] In some embodiments, the semi-solid ion-storing redox composition includes a solid including an ion-storage compound.

[0082] In some embodiments, the ion is proton or hydroxyl ion and the ion storage compound includes those used in a nickel-cadmium or nickel metal hydride battery.

[0083] In some embodiments, the ion is lithium and the ion storage compound is selected from the group consisting of metal fluorides such as CuF_2 , FeF_2 , FeF_3 , BiF_3 , CoF_2 , and NiF_2 .

[0084] In some embodiments, the ion is lithium and the ion storage compound is selected from the group consisting of metal oxides such as CoO, $\text{Co}_{0.3}\text{O}_{0.4}$, NiO, CuO, MnO.

[0085] In some embodiments, the ion is lithium and the ion storage compound includes an intercalation compound selected from compounds with formula $(\text{Li}_{1-x}\text{Z}_x)\text{MPO}_4$, where M is one or more of V, Cr, Mn, Fe, Co, and Ni, and Z is a non-alkali metal dopant such as one or more of Ti, Zr, Nb, Al, or Mg, and x ranges from 0.005 to 0.05.

[0086] In some embodiments, the ion is lithium and the ion storage compound includes an intercalation compound selected from compounds with formula LiMPO_4 , where M is one or more of V, Cr, Mn, Fe, Co, and Ni, in which the compound is optionally doped at the Li, M or O-sites.

[0087] In some embodiments, the ion is lithium and the ion storage compound includes an intercalation compound selected from the group consisting of

$\text{A}_x(\text{M}'_{1-a}\text{M}''_a)_y(\text{XD}_4)_z$,

$\text{A}_x(\text{M}'_{1-a}\text{M}''_a)_y(\text{DXD}_4)_z$, and

$\text{A}_x(\text{M}'_{1-a}\text{M}''_a)_y(\text{X}_{2\text{D}_7})_z$, wherein x, plus $y(1-a)$ times a formal valence or valences of M' , plus ya times a formal valence or valence of M'' , is equal to z times a formal valence of the XD_4 , X_{2D_7} , or DXD_4 group; and A is at least one of an alkali metal and hydrogen, M' is a first-row transition metal, X is at least one of phosphorus, sulfur, arsenic, molybdenum, and tungsten, M'' any of a Group IIA, IIIA, IVA, VA, VIA, VIIA, VIIIA, IB, IIB, IIIB, IVB, VB, and VIB metal, D is at least one of oxygen, nitrogen, carbon, or a halogen.

[0088] In some embodiments, the ion is lithium and the ion storage compound includes an intercalation compound selected from the group consisting of

$\text{A}_{1-a}\text{M}''_a\text{M}'_y(\text{XD}_4)_z$, $\text{A}_{1-a}\text{M}''_a\text{M}'_y(\text{DXD}_4)_z$ and $\text{A}_{1-a}\text{M}''_a\text{M}'_y(\text{X}_{2\text{D}_7})_z$, where $(1-a)x$ plus the quantity ax times the formal valence or valences of M'' plus y times the formal valence or valences of M' is equal to z times the formal valence of the XD_4 , X_{2D_7} or DXD_4 group, and A is at least one of an alkali metal and hydrogen, M' is a first-row transition metal, X is at least one of phosphorus, sulfur, arsenic, molybdenum, and tungsten, M'' any of a Group IIA, IIIA, IVA, VA, VIA, VIIA, VIIIA, IB, IIB, IIIB, IVB, VB, and VIB metal, D is at least one of oxygen, nitrogen, carbon, or a halogen.

[0089] In some embodiments, the ion is lithium and the ion storage compound includes an intercalation compound selected from the group consisting of ordered rocksalt compounds LiMO_2 including those having the including those having the $\alpha\text{-NaFeO}_2$ and orthorhombic- LiMnO_2 structure type or their derivatives of different crystal symmetry, atomic ordering, or partial substitution for the metals or oxygen, where M includes at least one first-row transition metal but may include non-transition metals Including but not limited to Al, Ca, Mg or Zr.

[0090] In some embodiments, the semi-solid ion storing redox composition includes a solid including amorphous carbon, disordered carbon, graphitic carbon, or a metal-coated or metal decorated carbon.

[0091] In some embodiments, the semi-solid ion storing redox composition can include a solid including nanostructures, e.g., nanowires, nanorods, and nanotetrapods.

[0092] In some embodiment, the semi-solid ion storing redox composition includes a solid including an organic redox compound.

[0093] In some embodiments, the positive electrode can include a semi-solid ion storing redox composition including a solid selected from the groups consisting of ordered rocksalt compounds LiMO_2 including those having the $\alpha\text{-NaFeO}_2$ and orthorhombic- LiMnO_2 structure type or their derivatives of different crystal symmetry, atomic ordering, or partial substitution for the metals or oxygen, wherein M Includes at least one first-row transition metal but may include non-transition metals including but not limited to Al, Ca, Mg, or Zr. The negative electrode can includes a semi-solid ion-storing composition including a solid selected from the group consisting of amorphous carbon, disordered carbon, graphitic carbon, or a metal-coated or metal-decorated

carbon.

[0094] In some embodiments, the positive electrode can include a flowable semi-solid ion-storing redox composition including a solid selected from the group consisting of

$A_{\text{sub}.x}(M'_{\text{sub}.1-aM''_{\text{sub}.a})_{\text{sub}.y}(XD_{\text{sub}.4})_{\text{sub}.z}$,

$A_{\text{sub}.x}(M'_{\text{sub}.1-aM''_{\text{sub}.a})_{\text{sub}.y}(DXD_{\text{sub}.4})_{\text{sub}.z}$, and

$A_{\text{sub}.x}(M'_{\text{sub}.1-aM''_{\text{sub}.a})_{\text{sub}.y}(X_{\text{sub}.2D_{\text{sub}.7})_{\text{sub}.z}$, and where x, plus y(1-a) times a formal valence or valences of M', plus ya times a formal valence or valence of M'', is equal to z times a formal valence of the XD_{sub.4}, X_{sub.2D.sub.7}, or DXD_{sub.4} group, and A is at least one of an alkali metal and hydrogen, M' is a first-row transition metal, X is at least one of phosphorus, sulfur, arsenic, molybdenum, and tungsten, M'' any of a Group IIA, IIIA, IVA, VA, VIA, VIIA, VIIIA, IB, IIB, IIIB, IVB, VB, and VIB metal, D is at least one of oxygen, nitrogen, carbon, or a halogen and the negative electrode includes a flowable semi-solid ion-storing redox composition including a solid selected from the group consisting of amorphous carbon, disordered carbon, graphitic carbon, or a metal-coated or metal-decorated carbon.

[0095] In some embodiments, the positive electrode can include a flowable semi-solid ion-storing redox composition including a compound with a spinel structure.

[0096] In some embodiments, the positive electrode includes a flowable semi-solid ion-storing redox composition including a compound selected from the group consisting of LiMn_{sub.2}O_{sub.4} and its derivatives; layered-spinel nanocomposites in which the structure includes nanoscopic regions having ordered rocksalt and spinel ordering; so-called “high voltage spinels” with a potential vs. Li/Li⁺ that exceeds 4.3V including but not limited to LiNi_{0.5}Mn_{1.5}O₄; olivines LiMPO_{sub.4} and their derivatives, in which M includes one or more of Mn, Fe, Co, or Ni, partially fluorinated compounds such as LiVPO_{sub.4}F, other “polyanion” compounds, and vanadium oxides V_{sub.x}O_{sub.y} including V_{sub.2}O_{sub.5} and V_{sub.6}O_{sub.11}.

[0097] In some embodiments the semi-solid battery is a lithium battery, and the anode **150** compound comprises graphite, graphitic or non-graphitic carbon, amorphous carbon, mesocarbon microbeads, boron-carbon alloys, hard or disordered carbon, lithium titanate spinel, or a solid metal or metal alloy or metalloid or metalloid alloy that reacts with lithium to form intermetallic compounds, e.g., Si, Ge, Sn, Bi, Zn, Ag, Al, any other suitable metal alloy, metalloid alloy or combination thereof. More particularly the metals, metal alloys, or metalloids are selected from a group that has a high capacity for intercalating lithium such as, for example, silicon, bismuth, barium, gallium, indium, zinc, tin, antimony, aluminum, gold, titanium oxide, molybdenum, germanium, manganese, niobium, vanadium, tantalum, iron, copper, chromium, nickel, cobalt, zirconium, yttrium, molybdenum oxide, germanium oxide, silicon oxide, silicon carbide, or a lithiated metal or metal alloy including such compounds as LiAl, Li_{sub.9}Al_{sub.4}, Li_{sub.3}Al, LiZn, LiAg, Li_{sub.10}Ag_{sub.3}, Li_{sub.5}B_{sub.4}, Li_{sub.7}B_{sub.6}, Li_{sub.12}Si_{sub.7}, Li_{sub.21}Si_{sub.8}, Li_{sub.13}Si_{sub.4}, Li_{sub.21}Si_{sub.5}, Li_{sub.5}Sn_{sub.2}, Li_{sub.13}Sn_{sub.5}, Li_{sub.7}Sn_{sub.2}, Li_{sub.22}Sn_{sub.5}, Li_{sub.2}Sb, Li_{sub.3}Sb, LiBi, or Li_{sub.3}Bi, or amorphous metal alloys of lithiated or non-lithiated compositions, any other high capacity materials or alloys thereof, or any other combination thereof. In some embodiments, the high capacity material in the anode **150** can result in the anode **150** having a capacity in the range of about 700 mAh/g to about 1,200 mAh/g.

[0098] In some embodiments, the electrochemical function of the semi-solid redox cell is improved by mixing or blending the cathode **140** and/or anode **150** particles with particulates of an electronically conductive material, such as solid inorganic conductive materials including but not limited to metals, metal carbides, metal nitrides, metal oxides, and allotropes of carbon including carbon black, graphitic carbon, carbon fibers, carbon microfibers, vapor-grown carbon fibers (VGCF), fullerenic carbons including “buckyballs”, carbon nanotubes (CNTs), multiwall carbon nanotubes (MWNTs), single wall carbon nanotubes (SWNTs), graphene sheets or aggregates of graphene sheets, and materials comprising fullerenic fragments. In some embodiments, such

electronically insulating organic redox compounds are rendered electronically active by mixing or blending with an electronically conductive polymer, including but not limited to polyaniline or polyacetylene based conductive polymers or poly(3,4-ethylenedioxythiophene) (PEDOT), polypyrrole, polythiophene, poly(p-phenylene), poly(triphenylene), polyazulene, polyfluorene, polynaphthalene, polyanthracene, polyfuran, polycarbazole, tetrathiafulvalene-substituted polystyrene, ferrocene-substituted polyethylene, carbazole-substituted polyethylene, polyoxyphenazine, polyacenes, or poly(heteroacenes).). In some embodiments, the resulting cathode or anode mixture has an electronic conductivity of at least about 10^{-6} S/cm. In other embodiments, the mixture has an electronic conductivity between approximately 10^{-6} S/cm and 10^{-3} S/cm. In other embodiments, the mixture has an electronic conductivity of at least about 10^{-5} S/cm, or at least about 10^{-4} S/cm, or at least about 10^{-3} S/cm, or at least about 10^{-2} S/cm or more.

[0099] In some embodiments, the particles included in the semi-solid anode or cathode can be configured to have a partial or full conductive coating.

[0100] In some embodiments, the semi-solid ion-storing redox composition includes an ion-storing solid coated with a conductive coating material. In some embodiments, the conductive coating material has higher electron conductivity than the solid. In some embodiments, the solid is graphite and the conductive coating material is a metal, metal carbide, metal oxide, metal nitride, or carbon. In some embodiments, the metal is copper.

[0101] In some embodiments, the solid of the semi-solid ion-storing material is coated with metal that is redox inert at the operating conditions of the redox energy storage device. In some embodiments, the solid of the semi-solid ion storing material is coated with copper to increase the conductivity of the storage material particle, to increase the net conductivity of the semi-solid, and/or to facilitate charge transfer between energy storage particles and conductive additives. In some embodiments, the storage material particle is coated with, about 1.5% by weight metallic copper. In some embodiments, the storage material particle is coated with about 3.0% by weight metallic copper. In some embodiments, the storage material particle is coated with about 8.5% by weight metallic copper. In some embodiments, the storage material particle is coated with about 10.0% by weight metallic copper. In some embodiments, the storage material particle is coated with about 15.0% by weight metallic copper. In some embodiments, the storage material particle is coated with about 20.0% by weight metallic copper.

[0102] In some embodiments, the conductive coating is placed on the cathode **140** and/or anode **150** particles by chemical precipitation of the conductive element and subsequent drying and/or calcination.

[0103] In some embodiments, the conductive coating is placed on the cathode **140** and/or anode **150** particles by electroplating (e.g., within a fluidized bed).

[0104] In some embodiments, the conductive coating is placed on the cathode **140** and/or anode **150** particles by co-sintering with a conductive compound and subsequent comminution.

[0105] In some embodiments, the electrochemically active particles have a continuous intraparticle conductive material or are bedded in a conductive matrix.

[0106] In some embodiments, a conductive coating and intraparticulate conductive network is produced by multicomponent-spray-drying, a semi-solid cathode **140** and/or anode **150** particles and conductive material particulates.

[0107] In some embodiments, the semi-solid composition also includes conductive polymers that provide an electronically conductive element. In some embodiments, the conductive polymers are or more of a polyacetylene, polyaniline, polythiophene, polypyrrole, poly(p-phenylene), poly(triphenylene), polyazulene, polyfluorene, polynaphthalene, polyanthracene, polyfuran, polycarbazole, polyacenes, poly(heteroacenes). In some embodiments, the conductive polymer is a compound that reacts in-situ to form a conductive polymer on the surface of the active material particles. In some embodiments, the compound can be 2-hexylthiophene or 3-hexylthiophene and

oxidizes during charging of the battery to form a conductive polymer coating on solid particles in the cathode semi-solid suspension. In other embodiments, redox active material can be embedded in conductive matrix. The redox active material can coat the exterior and interior interfaces in a flocculated or agglomerated particulate of conductive material. In some embodiments, the redox-active material and the conductive material can be two components of a composite particulate. Without being bound by any theory or mode of operation, such coatings can pacify the redox active particles and can help prevent undesirable reactions with carrier liquid or electrolyte. As such, it can serve as a synthetic solid-electrolyte interphase (SEI) layer.

[0108] In some embodiments, inexpensive iron compounds such as pyrite (FeS.sub.2) are used as inherently electronically conductive ion storage compounds. In some embodiments, the ion that is stored is Li⁺.

[0109] In some embodiments, redox mediators are added to the semi-solid to improve the rate of charge transfer within the semi-solid electrode. In some embodiments, this redox mediator is ferrocene or a ferrocene-containing polymer. In some embodiments, the redox mediator is one or more of tetrathiafulvalene-substituted polystyrene, ferrocene-substituted polyethylene, carbazole-substituted polyethylene.

[0110] In some embodiments, the surface conductivity or charge transfer resistance of current collectors **110/120** used in semi-solid battery is increased by coating the current collector surface with a conductive material. Such layers can also serve as a synthetic SEI layer. Non-limiting examples of conductive coating materials include carbon, a metal, metal-carbide, metal nitride, metal oxide, or conductive polymer. In some embodiments, the conductive polymer includes but is not limited to polyaniline or polyacetylene based conductive polymers or poly(3,4-ethylenedioxythiophene) (PEDOT), polypyrrole, polythiophene, poly(p-phenylene), poly(triphenylene), polyazulene, polyfluorene, polynaphthalene, polyanthracene, polyfuran, polycarbazole, tetrathiafulvalene-substituted polystyrene, ferrocene-substituted polyethylene, carbazole-substituted polyethylene, polyoxyphenazine, polyacenes, or poly(heteroacenes). In some embodiments, the conductive polymer is a compound that reacts in-situ to form a conductive polymer on the surface of the current collector. In some embodiments, the compound is 2-hexylthiophene and oxidizes at a high potential to form a conductive polymer coating on the current collector. In some embodiments; the current collector is coated with metal that is redox-inert at the operating conditions of the redox energy storage device

[0111] The semi-solid redox compositions can include various additives to improve the performance of the redox cell. The liquid phase of the semi-solids in such instances would comprise a solvent, in which is dissolved an electrolyte salt, and binders, thickeners, or other additives added to improve stability, reduce gas formation, improve SEI formation on the negative electrode particles, and the like. Examples of such additives included vinylene carbonate (VC), vinylethylene carbonate (VEC), fluoroethylene carbonate (FEC), or alkyl cinnamates, to provide a stable passivation layer on the anode or thin passivation layer on the oxide cathode, p ropene sultone (PS), propene sultone (PrS), or ethylene thiocarbonate as antigassing agents, biphenyl (BP), clohexylbenzene, or partially hydrogenated terphenyls, as gassing/safety/cathode polymerization agents, or lithium bis(oxalato)borate as an anode passivation agent.

[0112] In some embodiments, the semi-solid cathode **140** and/or anode **150** can include a non-aqueous liquid electrolyte that can include polar solvents such as, for example, alcohols or aprotic organic solvents. Numerous organic solvents have been proposed as the components of Li-ion battery electrolytes, notably a family of cyclic carbonate esters such as ethylene carbonate, propylene carbonate, butylene carbonate, and their chlorinated or fluorinated derivatives, and a family of acyclic dialkyl carbonate esters, such as dimethyl carbonate, diethyl carbonate, ethylmethyl carbonate, dipropyl carbonate, methyl propyl carbonate, ethyl propyl carbonate, dibutyl carbonate, butylmethyl carbonate, butylethyl carbonate and butylpropyl carbonate. Other solvents proposed as components of Li-ion battery electrolyte solutions include γ -butyrolactone,

dimethoxyethane, tetrahydrofuran, 2-methyl tetrahydrofuran, 1,3-dioxolane, 4-methyl-1,3-dioxolane, diethyl ether, sulfolane, methylsulfolane, acetonitrile, propionitrile, ethyl acetate, methyl propionate, ethyl propionate, dimethyl carbonate, tetraglyme, and the like. These nonaqueous solvents are typically used as multicomponent mixtures, into which a salt is dissolved to provide ionic conductivity, Exemplary salts to provide lithium conductivity include LiClO_4 , LiPF_6 , LiBF_4 , LiTFSI , LiBETI , LiBOB , and the like.

[0113] In some embodiments, the non-aqueous cathode **140** and/or anode **150** semi-solid compositions are prevented from absorbing impurity water and generating acid (such as HF in the case of LiPF_6 salt) by incorporating compounds that getter water into the active material suspension, or into the storage tanks or other plumbing of the system, for example, in the case of redox flow cell batteries. Optionally, the additives are basic oxides that neutralize the acid. Such compounds include but are not limited to silica gel, calcium sulfate (for example, the product known as Drierite), aluminum oxide and aluminum hydroxide.

[0114] In some embodiment, the cathode **140** can be a semi-solid cathode and the anode **150** can be a conventional anode for example, a solid anode formed from the calendering process as known as is commonly known in the arts. In some embodiments, the cathode **140** can be a semi-solid cathode and the anode **150** can also be a semi-solid anode as described herein. In some embodiments, the cathode **140** and the anode **150** can both be semi-solid flowable electrodes, for example, for use in a redox flow cell.

[0115] FIG. 2 shows a schematic illustration of an electrochemical cell **200** that includes a positive current collector **210**, a negative current collector **220**, and a separator **230** disposed between the positive current collector **210** and the negative current collector **220**. Both the positive current collector **210** and the negative current collector have a thickness t_3 and the separator **230** has a thickness t_4 . A cathode **240** having a thickness t_5 is disposed between the positive current collector **210** and the separator **230**, and an anode **250** having a thickness t_6 is disposed between the negative current collector **220** and the separator **230**.

[0116] The cathode **240** and anode **250** of the electrochemical cell **200** can be formed using a conventional coating and calendering process. For example, the cathode **240** and anode **250** can be manufactured by mixing electrochemically active ion-storage compounds (active materials), electronically conductive additives, and polymeric binders with a solvent to form a slurry, coating the slurry onto metal foil current collectors **210** and **220**, and then calendering the coated current collector under high pressure to increase density and control thickness of the finished electrode.

The finished electrode is then slit/cut into sizes and/or shapes that are appropriate for the form factor of the manufactured battery. The slit electrode composites can then be co-wound or co-stacked with intervening separator **230** layers to construct battery windings, i.e. “jelly-rolls” or “stacks”, which are then packaged in metal cans, flexible polymer pouches, etc. The resulting electrochemical cell is then infiltrated with liquid electrolyte in a carefully controlled environment.

[0117] The anode **250** can be a conventional anode (e.g., carbon-based) such that the anode **250** and the cathode **240** can have comparable capacities such as, for example, about 150-200 mAh/g. Thus, the thickness t_5 of the cathode **240** and the thickness t_6 of the anode **250** are substantially the same, e.g., about 150 μm , and the capacities of the anode **240** and the cathode **250** are “matched.” The thickness t_4 of the separator **220** is typically substantially less than the thickness t_5 of the cathode **240** and thickness t_6 of the anode **250** e.g., about 20 μm . The current collectors can be metal foils having a thickness comparable to the separator, e.g., about 20 μm .

[0118] One significant disadvantage of the conventional electrodes used in the electrochemical cell **200** is that the loading density of the active materials cannot be increased to increase the capacity of the electrodes, in part, due to manufacturing limitation of the coating/calendering process. Increasing the thickness of the electrodes in conventional electrochemical cells beyond 150-200 μm is also not practical because: (1) thicker electrodes made using the conventional

coating/calendering process tend to delaminate from the current collectors during the high speed coating process and rolling process, and (2) thick electrodes made using the conventional coating/calendering process have low conductivity, which dramatically increases the cell impedance. To overcome these limitations, a series of cells are typically stacked to obtain the desired capacity. This stacking of cells results in a significant portion of the volume of the battery to be occupied by inactive materials, i.e., the separator **230** and current collectors **210** and **220**. Therefore, the ratio of active to inactive materials is low, which correlates to a significant reduction in energy density.

[0119] For example, if the thickness $t_{sub.3}$ of the current collectors **210** and **220** is 20 μm each, and the thickness $t_{sub.4}$ of the separator **230** is also 20 μm , the total inactive material thickness is 60 μm . If the thickness $t_{sub.5}$ of the cathode **240** is 150 μm and the thickness $t_{sub.6}$ of the anode **250** is also 150 μm , the total active material thickness is 300 μm . Therefore, approximately 16% of the total thickness of the electrochemical cell **200** is occupied by inactive material, which does not contribute to the energy density and charge capacity of the battery.

[0120] One option for producing higher energy capacity is through the use of high capacity active materials, e.g., Si or Sn, in the anode composition. FIG. 3 shows a schematic illustration of a high capacity electrochemical cell **300** that includes a positive current collector **310**, a negative current collector **320**, and a separator **330** disposed between the positive current collector **310** and the negative current collector **320**. Both the positive current collector **310** and the negative (current collector **320** have a thickness $t_{sub.7}$ and the separator **330** has a thickness $t_{sub.8}$. A cathode **340** having a thickness $t_{sub.9}$ is disposed between the positive current collector **310** and the separator **330**, and a high capacity anode **350** having a thickness $t_{sub.10}$ is disposed between the negative current collector **320** and the separator **330**.

[0121] The cathode **340** can be any conventional cathode that uses binders. The cathode **340** can be manufactured using a conventional method as described with reference to FIG. 2. The thickness $t_{sub.9}$ of the conventional cathode **340** can be, for example, about 150 μm and can have a capacity, for example, of about 150-200 mAh/g. The anode **350** is a high capacity anode that includes a high capacity material, e.g., Si or Sn and can have any composition as described herein. The anode **350** can have a capacity of about 700-1,200 mAh/g. Although the anode **350** has a much higher capacity compared with a conventional anode, e.g., the anode **250**, the anode **350** capacity has to be matched with the capacity of cathode **350**. Since the conventional cathode **340** used in the electrochemical cell **300** cannot have a thickness $t_{sub.9}$ greater than about 150-200 μm due to limitations as described herein, the anode **350** has to be made thinner. For example, high capacity anode **350** can have a thickness $t_{sub.10}$ of about 30-50 μm to have a capacity that matches the capacity of the cathode **34** Coatings on the order of 30-50 μm begin to approach the thickness level of a single graphite particle and therefore are manufactured using expensive fabrication processes, e.g., atomic layer deposition, vapor deposition, sputtering, and the likes. This limitation prevents the full utilization of the potential energy capacity achievable with the high capacity anode **350**.

[0122] In order to increase the capacity of a battery, a series of individual cells **300** can be stacked, as described with reference to the electrochemical cell **200**. Although the capacity of electrochemical cells **200** and **300** are substantially equivalent, more individual cells **300** can into a "stack" of a given form factor because the anode **350** is significantly thinner than the anode **250**. However, a significant percentage of the finished battery is still occupied by the inactive materials. For example, the thickness $t_{sub.7}$ of the current collectors **310** and **320** is the same as current collectors **210** and **220** (i.e., 20 μm each), and the thickness of the separator **330** is the same as the separator **230** (i.e., 20 μm), which represents the total inactive material thickness of 60 μm . If the thickness $t_{sub.9}$ of the cathode **340** is the same as the cathode **240** (i.e., 150 μm) and the thickness $t_{sub.10}$ of the high capacity anode **350** is 50 μm , the total active material thickness is 200 μm . Therefore, approximately 23% of the total thickness of the electrochemical cell **300** is occupied by inactive material, which does not contribute to energy density and charge capacity of the battery,

Although the capacity of the finished battery in this example utilizing electrochemical cells **300** would be greater than the capacity of the finished battery utilizing electrochemical cells **200** because the thinner anodes **350** allow more cells **300** to fit into the same form factor, a higher percentage (23% vs. 16%) of the finished battery is made from materials that are not contributing to charge capacity of the battery, thus, reducing its energy density. Said another way, while the high capacity anode **350** enables the stacking of more electrochemical cells **300** in a battery compared to the conventional anode **250** to get a higher capacity battery, the percentage of the inactive material is also higher which negates some of this benefit.

[0123] As described herein, increasing the thickness of the cathode and/or increasing the active material loading in the cathode are two ways that the benefits of a high capacity anode can be utilized. While these are not viable options with conventional cathodes for the reasons described above, various embodiments of semi-solid cathodes described herein can be made thicker than the maximum thickness of 200 μm achievable with conventional cathodes and with loading densities that can be up to 5 times higher than conventional cathodes. FIG. 4 shows a schematic illustration of an electrochemical cell **400** that includes a positive current collector **410**, a negative current collector **420**, and a separator **430** disposed between the positive current collector **410** and the negative current collector **420**. Both the positive current collector **410** and the negative current collector **420** have a thickness $t_{\text{sub.11}}$ and the separator **430** has a thickness $t_{\text{sub.12}}$. A semi-solid cathode **440** having a thickness $t_{\text{sub.13}}$ is disposed between the positive current collector **410** and the separator **430**, and a high capacity anode **450** having a thickness $t_{\text{sub.14}}$ is disposed between the negative current collector **420** and the separator **430**.

[0124] The anode **450** is a high capacity anode that can be formed using conventional methods as described before with reference to FIG. 2 or any other conventional method. In some embodiments, the anode **450** can be a stationary semi-solid or a flowable semi-solid anode as described herein. The cathode **440** is a semi-solid cathode that can be a stationary semi-solid cathode or a flowable semi-solid cathode as described herein. In some embodiments, the semi-solid cathode is substantially free from binders that are typically used in conventional calendered electrodes. In contrast to conventional cathodes, e.g., the cathodes **240** and **340**, the semi-solid cathode **440** can be made thicker than 200 μm such as, for example, thicker than 250 μm , 300 μm , 350 μm , 400 μm , 450 μm , 500 μm , 600 μm , 700 μm , 800 μm , 900 μm , 1,000 or, 500 μm , and up to 2,000 μm or more as described herein. In some embodiment, the cathode can be at least about two times, three times, four times, five times, or even six times the thickness of the anode. The semi-solid cathode **440** can also formulated to have up to five times the loading density of conventional cathodes. The thicker cathode **440** and higher loading densities can be achieved with the semi-solid cathode **440** without the negative consequences associated with thicker conventional electrodes.

[0125] The ability to substantially increase the thickness $t_{\text{sub.13}}$ of the semi-solid cathode **440** eliminates the restriction on the thickness $t_{\text{sub.14}}$ of the high capacity anode **450** that is imposed by the limitations of the conventional cathodes, e.g., cathode **250** and **350**. The anode **450** can therefore be made thicker to match the capacity of the thicker cathode **450**. For example, the anode **450** can a thickness $t_{\text{sub.14}}$ of about 70 μm to about 200 μm using conventional coating/calendering manufacturing methods or upwards of 600 μm or even higher if a semi-solid anode is used as describe herein.

[0126] Therefore, the semi-solid cathode **440** and high capacity anode **450** of the electrochemical cell **400** formed using formulations and methodologies described herein can allow the maximum capacity offered by the high capacity materials of the anode **450** to be utilized in an electrochemical cell with fewer stacks. In some embodiments, the semi-solid cathode **440** and high capacity anode **450** can be used to make a single electrochemical cell **400** without any stacking. Because stacking is not needed or the number of stacks in a battery using the semi-solid cathode **440** is substantially less than the number of stacks needed in a high capacity anode and conventional cathode battery, for example electrochemical cell **300**, or a conventional anode, conventional cathode battery, for

example, electrochemical cell **200**, the ratio of active materials to inactive materials in a cell stack formed from electrochemical cell **400**, is substantially higher than that of the electrochemical cells **200** and **300**. Thus, most of the volume of the assembled battery is occupied by active charge storing materials, which leads to increased overall charge capacity and energy density of the battery. For example, as shown in FIG. 4, the thickness $t_{sub.11}$ of the current collectors **410** and **420** is the same as current collectors **310** and **320** (i.e., 20 μm each), and the thickness $t_{sub.12}$ of the separator **430** is the same as the separator **330** (i.e., 20 μm), which represents the total inactive material thickness of 60 μm . If the thickness $t_{sub.13}$ of the semi-solid cathode **440** is 500 μm and the thickness $t_{sub.14}$ of the high capacity anode **450** is 150 μm , the total active material thickness is 650 μm . Therefore, approximately 8.4% of the total thickness of the electrochemical cell **400** is occupied by inactive material, which does not contribute to the energy density and charge capacity of the battery. Since the percentage of inactive materials in the electrochemical cell **400** is approximately half the percentage of inactive materials in the electrochemical cell **200** (16%) and approximately one-third the percentage of inactive materials in the electrochemical cell **300** (23%), and because the full utilization of the high capacity anode **450** and higher loading density of the semi-solid cathode **440**, the electrochemical cell **400** has a much higher energy density than the other electrochemical cells **200** and **300**.

[0127] Furthermore, by utilizing a semi-solid anode, the high capacity anode **450** can also be made thicker than 150 μm and the semi-solid cathode **440** can be made thicker than 500 μm to match the thicker high capacity anode **450** further reducing the percentage of inactive materials, which do not contribute to the energy density and charge capacity of the battery. For example, if the thickness $t_{sub.14}$ of a semi-solid high capacity anode **450** is 600 μm and the thickness $t_{sub.13}$ of the semi-solid cathode **440** is increased to 2,000 μm , the total active material thickness is 2,600 μm . If the thickness of the current collectors **410** and **420**, and separator **430** have the same thickness as above (i.e., of the 20 μm each), the percentage of inactive materials is approximately only 2.2%. Therefore, electrochemical cells **400** made with the aforesaid materials and methods have a substantially higher commercial appeal due, in part, to their increased energy density and decreased cost.

[0128] While various embodiments of the system, methods and devices have been described above, it should be understood that they have been presented by way of example only, and not limitation, Where methods and steps described above indicate certain events occurring in certain order, those of ordinary skill in the art having the benefit of this disclosure would recognize that the ordering of certain steps may be modified and such modification are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. The embodiments have been particularly shown and described, but it will be understood that various changes in form and details may be made.

Claims

1-29. (canceled)

30. An electrochemical cell, comprising: a semi-solid electrode; a high capacity electrode; and an interposing layer disposed between the high capacity electrode and the semi-solid electrode, wherein the semi-solid electrode has a thickness of at least about two times a thickness of the high capacity electrode.

31. The electrochemical cell of claim 30, wherein the semi-solid electrode includes a slurry of a mixture of an active material and a conductive material in a liquid electrolyte.

32. The electrochemical cell of claim 30, wherein the thickness of the semi-solid electrode is in a range of about three to about six times the thickness of the high capacity electrode.

33. The electrochemical cell of claim 30, wherein the high capacity electrode includes a lithium-

alloy-forming compound including at least one of tin, tin alloys, silicon, silicon alloys, antimony, aluminum, or titanium oxide.

34. The electrochemical cell of claim 30, wherein the high capacity electrode includes a mixture of an active material in a liquid electrolyte.

35. The electrochemical cell of claim 34, wherein the liquid electrolyte is a non-aqueous liquid electrolyte.

36. The electrochemical cell of claim 34, wherein the high capacity electrode includes about 40% to about 75% by volume of the active material.

37. The electrochemical cell of claim 30, wherein the high capacity electrode has a thickness in a range of about 60 μm to about 150 μm .

38. The electrochemical cell of claim 30, wherein the high capacity electrode has a thickness in a range of about 30 μm to about 600 μm .

39. An electrochemical cell, comprising: a first electrode, the first electrode including a mixture of an active material and a conductive material in a liquid electrolyte; a second electrode including a high capacity material; and an interposing layer disposed between the first electrode and the second electrode, wherein the second electrode has a thickness of less than about 50% of the thickness of the first electrode.

40. The electrochemical cell of claim 39, wherein the second electrode includes a lithium-alloy-forming compound including at least one of tin, tin alloys, silicon, silicon alloys, antimony, aluminum, or titanium oxide.

41. The electrochemical cell of claim 40, wherein the lithium-alloy-forming compound includes silicon.

42. The electrochemical cell of claim 40, wherein the lithium-alloy-forming compound includes tin.

43. The electrochemical cell of claim 39, wherein the thickness of the first electrode is in a range of about 3 times to about 6 times the thickness of the second electrode.

44. The electrochemical cell of claim 39, wherein the second electrode is a semi-solid anode including a mixture of an anode active material in a liquid electrolyte.

45. The electrochemical cell of claim 44, wherein the liquid electrolyte in the semi-solid anode is a non-aqueous liquid electrolyte.

46. The electrochemical cell of claim 39, wherein the second electrode has a thickness in a range of about 30 μm to about 600 μm .

47. An electrochemical cell, comprising: a first semi-solid electrode having a first thickness; a second semi-solid electrode including a high capacity material, the second semi-solid electrode having a second thickness in a range of about 20% to about 50% of the first thickness; and an interposing layer disposed between the first semi-solid electrode and the second semi-solid electrode.

48. The electrochemical cell of claim 47, wherein the thickness of the first semi-solid electrode is at least about two times the thickness of the second semi-solid electrode.

49. The electrochemical cell of claim 47, wherein the second semi-solid electrode is a semi-solid anode including about 40% to about 75% by volume of an active material.

50. The electrochemical cell of claim 47, wherein the second semi-solid electrode has a thickness in a range of about 30 μm to about 600 μm .

51. The electrochemical cell of claim 47, wherein the first semi-solid electrode has a thickness in a range of about 250 μm to about 2,000 μm .

52. The electrochemical cell of claim 47, wherein the second semi-solid electrode is a semi-solid anode including a carbonaceous active material, a high energy capacity anode material, and a conductive material in a liquid electrolyte.

53. The electrochemical cell of claim 47, wherein the second semi-solid electrode includes a lithium-alloy-forming compound including at least one of tin, tin alloys, silicon, silicon alloys,

antimony, aluminum, or titanium oxide.

54. The electrochemical cell of claim 53, wherein the lithium-alloy-forming compound includes silicon.

55. The electrochemical cell of claim 53, wherein the lithium-alloy-forming compound includes tin.
